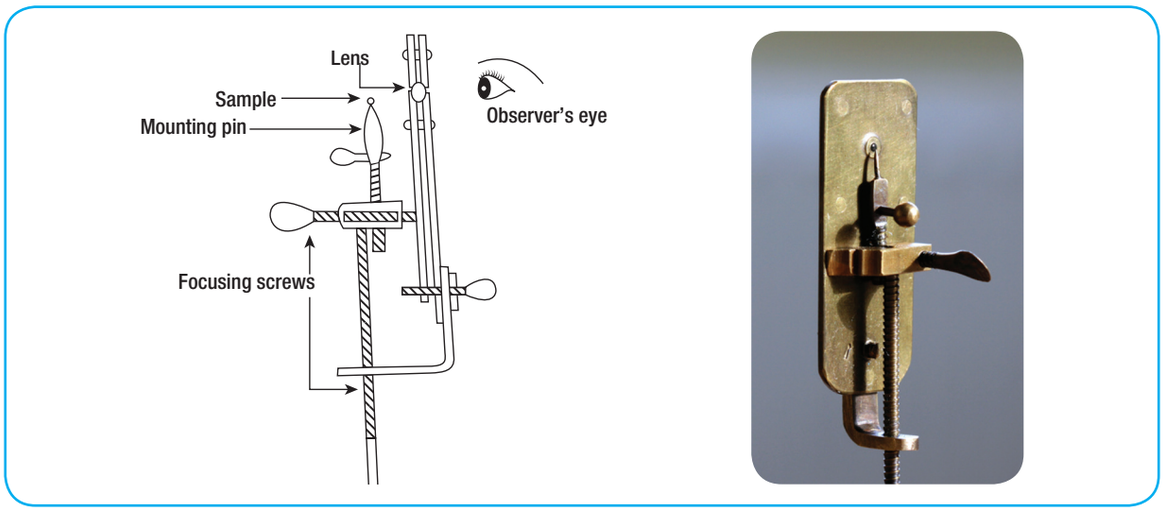


Figure 1.2: Leeuwenhoek's Microscope



Figure 1.3: Louis Pasteur (1822-1895)

Contribution to Microbiology

To wine industry

Louis Pasteur discovered alcohol production from grape juice was due to yeast. The presence or contamination of rod shaped bacteria resulted in large amounts of lactic acid production in wine. He also found that microorganisms in fermented fruits and grains, resulting in alcohol production. He coined the term "**fermentation**".

Pasteur disproved spontaneous generation

Spontaneous generation states that life could arise spontaneously from inanimate (non-living) materials (Abiogenesis). Pasteur disproved the theory of spontaneous generation. He strongly supported theory of Biogenesis (life orginates from pre-existing life forms). To prove this he carried out several experiments. Pasteur poured meat infusions into flasks and then drew the top of each flask into a long curved neck that would admit air but not dust (Figure 1.4). He found that if the infusions were heated, they remained sterile (free from any growth) until they were exposed to dust. After opening them on a dusty road and resealing them, he demonstrated the growth of microorganisms in all the flasks. The unopened flasks were sterile. Thus he disproved the theory of spontaneous generation.

Pasteurization

Louis Pasteur used heat to destroy undesirable microbes in fruit juices. He employed 62.8°C (145°F) for 30 mins to kill microbes. This process is called Pasteurization which is commonly used in distillaries and dairy industry.

Discovery of diseases

Louis Pasteur found that Pebrine disease in silk worm was caused by a protozoan parasite. He suggested that Pebrine disease could be eliminated by using only healthy,

TN_GOVT_XI_Micro_Biology_CH01.indd 4 16-02-2019 9.54.22 AM